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Project Title:	Structural Chromosome Aberrations Formed in Response to Changes in Proton Energy and Dose Rate		
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Zip Code:	77555-5302	**Congressional District:**	14
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	From the perspective of NASA, risk uncertainties related to radiation exposure during space exploration are unacceptably large. These include the effects of sporadic solar particle events (SPEs) which can produce relatively high levels of radiation, principally from protons. The resultant dose to personnel from an SPE can be high enough to warrant concerns of cancer induction. Because human epidemiological data in this area are not available, reduction in risk uncertainties will require data from meaningful surrogate endpoints, such as chromosome damage. A reasonable and widely held notion is that dose rate effects from protons will be the same as for x- and gamma rays. There is, however, not much direct experimental evidence to support this contention, which adds additional uncertainty to risk projections. We propose to simulate an SPE using proton emissions from a particle accelerator at Loma Linda University. Owing to differences in ionization density that may translate to differences in biological effect, proton energies of 50 MeV and 250 MeV will be studied. These will be delivered at both high and low dose rates. Results will then be compared to		

Task Description:	those obtained for gamma rays from a 137-Cesium source. Much is known about chromosomal response to other types of radiation, such as x- and gamma rays, including how the level of damage is influenced by the rate at which radiation is delivered. We have chosen structural damage to chromosomes of human cells in culture as a surrogate endpoint for cell death and cancer. We first propose to construct quantitative dose responses for the induction of chromosome aberrations found in cells when they arrive at the first mitosis following exposure. For this and other objectives, multiplex fluorescence in situ hybridization (mFISH) will be used, which effectively paints each of the 23 pairs of human chromosomes a unique color. This allows even very complicated rearrangements among all chromosomes to be visualized simultaneously. The second set of objectives involves analysis of chromosome damage in clonal descendents of single cells that survive exposure to radiation. Several such clones will be expanded in culture and examined by mFISH for transmissible chromosome aberrations. A subset of these clones will be further examined for structural aberrations that are too small to be seen under the microscope, by employing comparative genomic hybridization applied to DNA microarrays (aCGH). It is argued that genetic damage in cells surviving radiation may be diagnostic of the type of radiation encountered, and that such damage has more direct relevance to carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overarching objective of these studies is to help reduce uncertainties associated with exposure to ionizing radiation in a space environment, specially those associated with exposure to solar particle events (SPEs) that relate chiefly to high energy proton exposure. Currently the assumption is that such protons behave essentially like other forms of low LET (linear energy transfer) photons such as X- and gamma rays, for which much quantitative information already exists. The extent to which the clastogenic (chromosome aberration producing) effects of protons actually are (or are not) similar to photons in the context of energy and dose rate is an important piece of information needed to help reduce the aforementioned uncertainies.
Task Progress:	Our objectives relate to quantifying the clastogenic effects of changes in instantaneous dose rates for energetic protons, compared to photons emitted from 137Cs. Our provisional plan was to use both 50 MeV and 250 MeV protons from Loma Linda Hospital, and compare these radiations, under conditions of both acute and chronic dose rates, to 662 keV photons. While we would have preferred to examine the effects of photons of energy >250 MeV, this represents the maximum (and minimum) proton energies available at this facility. However, since the time that our proposal was funded, the NSRL has invested considerable effort in making lower energies available, making uniform field sizes conveniently larger, and anticipating the longer runs necessary to accommodate researchers, such as us, who require moderately high doses to be delivered at very low dose rates (i.e., long exposure times). Consequently, we have decided instead to use the NSRL beam line at Brookhaven National Laboratory, and are putting in for runs in the spring and fall of 2009. In the mean time, we have been modeling the effects of chromosome aberration induction as a function of low LET dose (D) and dose rate (Dr), using the G-function, G(x), which returns multiplicative values (from zero to unity) on the interactive (aD^2) term. The predicted yield of aberrations (Y) is given by the expression Y=alpha D + G(x) • (beta D^2)+C; where alpha and beta are coefficients, and c is a constant representing the background frequency of aberrations. Initial modeling of this type is important, because it serves to guide experiments that, in all likelihood, will be resource-intensive in terms of total doses and dose rates that are likely to show a diminution of effect at chronic doses (i.e. total exposure time). We started with the premise that there is little point in studying dose rate effects for protons under conditions that are not first shown to occur for protons, since the underlying question for NASA is whether, and to what extent, protons behave like photons in terms of dose rate effects. Coefficients were derived from 24-color mFISH studies conducted in our laboratory for acute dose rates from 137Cs. The rejoining rate constant, expressed in terms of half-time, was derived from studies using premature chromosome condensation (PCC). From such analyses, we conclude that for total doses below about 2 Gy, it would be difficult to distinguish experimentally the effects of chronic dose rate delivery from that of acute dose delivery for low dose rates above about 0.5 cGy/min. And even at 0.5 cGy/min it would be desirable--for the purposes of quantitatively characterizing the shape of the response--to deliver total doses to include those in the 3-4 Gy range. Consequently, we would need, for a continuous exposure to protons, somewhere between 10-13 hours of proton beam time. Experimental systems with dose responses showing more curvature when delivered acutely (i.e., lower alpha/beta ratios) will lead to minimal dose estimates that are lower, hence requiring less beam time. Interestingly, the above figure also predicts that all dose rates (within experimental limits of measurement) above about 5 cGy/min can be considered acute exposures. We have the capability in our facility to irradiation cells at a nominal dose rate of 5 cGy/min with gamma rays, and are currently testing this assertion experimentally. Experiments using lower dose rates (e.g., 0.5 cGy/min) of gamma rays at Colorado State University are also underway. It may be possible to limit dose rates (hence reducing overall exposure times) to < or =2 Gy, but that will require careful scoring of many cells, the actual number depending, to a large degree, on the background frequency of aberrations in unirradiated cells. This relates to Objective 1 for which bulk populations of irradiated cells will be scored at the first mitosis following irradiation, and for Objective 2 of the proposal, in which randomly selected clonal derivatives of hTERT-immortalized human fibroblasts demonstrating long-term survival to radiation are to be assessed. We have amassed data for a population of cells that were transported to the NSRL facility, received no dose, and were than shipped back to our laboratory in Galveston for subcloning. Seven independent clones were then isolated, and multiple cells were scored from each by mFISH. Results show that the background frequency of aberrations, specifically translocations was remarkable low; some 96% of all cells scored showed a normal 46,XY karyotype, and only 0.6% contained translocations. NASA also jointly funds a project with DOE which involves a comparison of chromosome damage in cells irradiated in vitro compared to that caused by irradiation in 3D tissue constructs. We have chosen to use an artificial skin model which, for this purpose —and for the purposes of consolidating approaches and resources—are also considering for use in the current proposal. This model mimics the essential characteristics of natural skin, including cellular functions of epithelial and dermal layers. The model differs from that commercially available from various sources in that is was constructed using hTERT-immortalized fibroblasts in the dermal compartment. Immortalization allows for the recovery,

	and clonal expansion, of cells surviving radiation, without the problems normally otherwise associated with cell senescence of fibroblasts. For that reason it is ideal for use in Objectives 2 and 3 of this proposal, where extensive clonal expansion of cells surviving radiation is required. To date we have been successfully in incorporating hTERT fibroblasts into the dermal layer, and have shown that dermal fibroblasts can be recovered from such a 3D tissue constructs for the purposes of karyotypic analyses. Progress has been somewhat hampered during this first period of the grant because of our decision to employ protons from NSRL rather than Loma Linda, and because work is still underway to make a stable wide-field proton beam available to NASA researchers for low dose rate work. Collaborations with investigators at Colorado State University (Bedford) are ongoing in order to consolidate NSRL resources related to low dose-rate proton exposures. Karyotypic analysis has been temporarily placed on hold because the older mFISH analysis system we have has recently been giving us problems. Dedicated funds from the current grant were used to purchase a new mFISH system (Metasystems) and that equipment is in the process of being set up.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
Abstracts for Journals and Proceedings	Loucas BD, Cornforth MN. "Characterization of chromosome damage in dermal fibroblasts irradiated in vitro Versus in situ in a 3D human skin model." 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, June 30-July 2, 2008. Abstracts, 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, June 30-July 2, 2008. , Jun-2008
Abstracts for Journals and Proceedings	Cornforth MN. "Invited talk: Chromosome Damage as a Function of LET, Dose and Dose Rate and Its Role in Risk from Radiation Exposure." Low Dose and Low dose rate Radiation Effects and Models. 44th Annual Meeting of the National Council on Radiation Protection and Measurements (NCRP), Bethesda, MD, April 14-15, 2008. Low Dose and Low dose rate Radiation Effects and Models. 44th Annual Meeting of the National Council on Radiation Protection and Measurements (NCRP), Bethesda, MD, April 14-15, 2008. , Apr-2008
Articles in Peer-reviewed Journals	Bailey SM, Cornforth MN. "Telomeres and DNA double-strand breaks: Ever the twain shall meet?" Cellular and Molecular Life Sciences 2007 Nov;64(22):2956-64. PMID: 17876526 , Nov-2007
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